ProbMeTTa: ProbLog in MeTTa

Formalization, Implementation, and Verified Correctness

Abstract

We present ProbMeTTa, an implementation of ProbLog's distribution semantics in the MeTTa programming language, running on the PeTTa (Prolog-backed MeTTa) runtime. We formalize ProbLog's distribution semantics in Lean 4 and prove that ProbMeTTa's BDD-based inference correctly computes ProbLog query probabilities. The formalization comprises approximately 1,160 lines of Lean with zero unproven assumptions (`sorry`), covering BDD construction correctness, Weighted Model Counting, and compilation soundness and completeness. All 27 ProbMeTTa test cases pass, and 7 representative programs are additionally verified by kernel-checked conformance tests in Lean.

1 Introduction

ProbLog [1] is a probabilistic extension of Prolog based on Sato's distribution semantics [2]. It combines the declarative expressiveness of logic programming with principled probabilistic reasoning, and has been applied to link prediction, bioinformatics, and probabilistic databases.

MeTTa is a functional-relational programming language designed for AI reasoning. PeTTa is a Prolog-backed MeTTa implementation that compiles MeTTa expressions to SWI-Prolog.

ProbMeTTa implements ProbLog's exact inference engine in MeTTa using Binary Decision Diagrams (BDDs) for knowledge compilation and Weighted Model Counting (WMC) for probability computation, following the approach of Fierens et al. [3].

This paper presents:

1. A self-contained formalization of ProbLog's distribution semantics in Lean 4
2. ProbMeTTa: a working ProbLog implementation in MeTTa (27/27 tests passing)
3. A formal proof that ProbMeTTa correctly computes ProbLog probabilities
4. Kernel-checked conformance tests linking the Lean formalization to the MeTTa code

2 ProbLog: Distribution Semantics

2.1 Programs

A ProbLog program P consists of:

- **Probabilistic facts**: $p_1 :: f_1,\ \ldots,\ p_n :: f_n$ — each ground atom f_i is independently true with probability $p_i \in [0, 1]$.
- **Definite clauses**: $head \leftarrow body_1, \ldots, body_k$ — standard Horn clauses.

Example 2.1 (Alarm Network)**.**
```
0.1 :: burglary.
0.2 :: earthquake.
alarm :- burglary.
alarm :- earthquake.
query(alarm).
```

2.2 Semantics

ProbLog defines query probability via the *distribution semantics* [2]:

Definition 2.1 (Total Choice)**.** A *total choice* is an assignment $a : \{1, \ldots, n\} \to \{\texttt{true}, \texttt{false}\}$ that independently sets each probabilistic fact to true or false.

Definition 2.2 (Residual Program)**.** For total choice a, the *residual program* consists of the original rules plus the probabilistic facts chosen true: $KB(a) = \mathit{rules} \cup \{f_i \mid a(i) = \texttt{true}\}$.

Definition 2.3 (Query Probability)**.** Query q *holds under* a iff q is in the least Herbrand model of $KB(a)$. The *query probability* is:

$$P(q) = \sum_a w(a) \cdot \mathbf{1}[q \text{ holds under } a]$$

where $w(a) = \prod_{i=1}^{n} \big(a(i) \cdot p_i + (1 - a(i)) \cdot (1 - p_i)\big)$.

For the alarm example: $P(\mathit{alarm}) = 1 - (1 - 0.1)(1 - 0.2) = 0.28$.

3 BDD-Based Inference

ProbLog computes query probabilities via knowledge compilation to Binary Decision Diagrams (BDDs) [4], following Fierens et al. [3].

3.1 Binary Decision Diagrams

A BDD over n Boolean variables is a rooted directed acyclic graph with:

- Two terminal nodes: **0** (false) and **1** (true)
- Internal nodes labeled by variable $v \in \{1, \ldots, n\}$, each with a *lo* child (variable false) and *hi* child (variable true)

A Reduced Ordered BDD (ROBDD) additionally satisfies:

1. **Ordered**: variables appear in consistent order on all paths
2. **Reduced**: no node has $lo = hi$ (elimination rule)

3.2 Weighted Model Counting

Given a BDD f and probability weights $p_1, \ldots, p_n$:

$$\mathrm{WMC}(\mathbf{0}) = 0 \tag{1}$$

$$\mathrm{WMC}(\mathbf{1}) = 1 \tag{2}$$

$$\mathrm{WMC}(\mathrm{node}(v, lo, hi)) = (1 - p_v) \cdot \mathrm{WMC}(lo) + p_v \cdot \mathrm{WMC}(hi) \tag{3}$$

Theorem 3.1 (WMC Correctness)**.** For any ROBDD f representing Boolean function φ:

$$\mathrm{WMC}(f) = \sum_{\substack{a \in \{0,1\}^n \\ \varphi(a)=1}} \prod_{i=1}^{n} \big(a_i \cdot p_i + (1 - a_i)(1 - p_i)\big)$$

This is exactly the distribution semantics probability when φ is the query's success predicate.

3.3 Compilation

ProbLog compiles queries to BDDs by proof enumeration:

1. Each probabilistic fact f_i becomes BDD variable i
2. A conjunction $(b_1, \ldots, b_k)$ compiles to AND of sub-BDDs
3. Multiple derivations of the same head are OR'd together

4 ProbMeTTa: Implementation in MeTTa

ProbMeTTa implements the above algorithm in two MeTTa libraries:

4.1 `lib_bdd.metta`: BDD Operations

```
;; BDD node: (bdd-node $id $var $lo $hi)
;; Terminals: bdd-0 (false), bdd-1 (true)

;; Make node with elimination rule
(= (mk $var $lo $hi)
    (if (== $lo $hi) $lo
        (case (once (match &bdd ...))
            ((Empty (let $id (new-id) ...))
             ($id $id)))))

;; Shannon expansion with memoization
(= (apply-bdd $op $f $g) ...)

;; Weighted Model Counting
(= (bdd-wmc $f $env)
    (if (is-terminal $f)
        (if (== $f bdd-1) 1.0 0.0)
        (let* (($var (bdd-var $f))
               ($val (env-lookup $var $env)))
            (+ (* (- 1 $val) (bdd-wmc (bdd-lo $f) $env))
               (* $val (bdd-wmc (bdd-hi $f) $env))))))
```

4.2 `lib_prob.metta`: Probabilistic Inference

```
;; Declare probabilistic fact: p :: f
(= (:: $prob $goal)
    (progn (add-atom &prob-templates (prob-template $prob $goal))
           (add-atom &self (rule $goal (pf $goal)))))

;; Declare definite clause: head :- body
(= (prob-rule $body $head)
    (let $goals (body-goals $body)
        (add-atom &self (rule $head $goals))))

;; Resolve a goal, threading BDD trace
(= (prob-goal $goal $trace)
    (let $body (match &self (rule $goal $b) $b)
        (case $body
            (((pf $atom) (prob-assume $atom $trace))
             ($goals (exec-conj $goals $trace))))))
```

```
;; Top-level query
(= (?prob $goal)
   (let* (($final-bdd (?prob-bdd $goal))
          ($env (prob-wmc-env)))
      (bdd-wmc $final-bdd $env)))
```

4.3 Test Results

ProbMeTTa passes all 27 tests, covering:

Test	ProbLog Feature	Expected
Alarm network	Noisy-OR, 3-hop chain	$P = 0.196$
Fever chain	6 rules, shared derivations	$P = 0.35168$
Simple negation	Negation-as-failure	$P = 0.7$
Conditioning	$P(A \mid B) = P(A \wedge B)/P(B)$	$P = 0.357$
Annotated disjunctions	Multi-valued random variables	$P = 0.6$
Graph reachability	Recursive probabilistic rules	$P = 0.258$
Overlapping rules	Multiple derivation paths	$P = 0.49$

Table 1: Representative ProbMeTTa test results (all 27 pass).

5 Lean Formalization

The formalization is in Lean 4 (v4.28.0 with Mathlib v4.28.0), totaling approximately 1,160 lines across 6 files with zero `sorry` (unproven assumptions).

5.1 BDD Core (`Core.lean`, 169 lines)

```
inductive BDD (n : N) where
  | zero : BDD n                              -- false
  | one  : BDD n                              -- true
  | node (v : Fin n) (lo hi : BDD n) : BDD n -- if v then hi else lo

def BDD.eval (f : BDD n) (env : Fin n -> Bool) : Bool
```

The denotational semantics `BDD.eval` maps each BDD to the Boolean function $(a_1, \ldots, a_n) \mapsto \{0, 1\}$ it represents.

5.2 WMC Correctness (`WMC.lean`, 239 lines)

Theorem 5.1 (`bdd_wmc_correct`)**.** For any well-formed BDD f with probability weights $p_i \leq 1$:

$$\texttt{bdd\_wmc}(f, env) = \sum_{a \in \{0,1\}^n} \mathbf{1}[f.eval(a)] \cdot \prod_{i=1}^{n} w(i, a_i)$$

where $w(i, \texttt{true}) = p_i$ and $w(i, \texttt{false}) = 1 - p_i$.

The proof proceeds by structural induction on the BDD, with the key lemma being the *Shannon decomposition* for weighted sums: splitting the sum by one variable's value and factoring out its weight.

5.3 Operations Correctness (`Operations.lean`, 245 lines)

Theorem 5.2 (`apply_eval`)**.**

$$(\texttt{apply}\ op\ f\ g).eval(a) = op(f.eval(a),\ g.eval(a))$$

This proves that BDD construction via Shannon expansion preserves Boolean semantics. Seven conformance tests verify the Lean BDD operations against ProbMeTTa's expected results:

Program	Boolean function	Verification
Alarm	$v_0 \vee v_1$	`by simp`
Conjunction	$v_0 \wedge v_1$	`by simp`
Negation	$\neg v_0$	`by simp`
Neg. conjunction	$v_0 \wedge \neg v_1$	`by simp`
Overlapping	$(v_0 \wedge v_1) \vee v_2$	`by simp`
Fever chain	$(v_2 \wedge v_3) \vee ((v_0 \vee v_1) \wedge v_2 \wedge v_4)$	`by simp`
Calls-mary	$(v_0 \vee v_1) \wedge v_2$	`by simp`

Table 2: Kernel-checked conformance tests. Each `by simp` is verified by Lean's kernel.

5.4 Compilation Correctness (`Compilation.lean`, 268 lines)

The compilation relation `GroundBDDCompile` mirrors ProbMeTTa's `prob-goal` procedure:

Constructor	ProbMeTTa function	Meaning
`.fact`	`prob-assume`	Probabilistic fact → BDD variable
`.ruleG`	`prob-goal + exec-conj`	Rule application → AND of body BDDs
`.disj`	`bdd-disjunction`	Multiple derivations → OR

Theorem 5.3 (Soundness: `GroundBDDCompile_sound`)**.** If `GroundBDDCompile prog q f` and $f.eval(a) = \texttt{true}$, then $q \in \mathrm{LHM}(KB(a))$.

Theorem 5.4 (Completeness: `GroundBDDCompile_complete`)**.** If $q \in \mathrm{LHM}(KB(a))$, then there exists f such that `GroundBDDCompile prog q f` and $f.eval(a) = \texttt{true}$.

Completeness is proved by induction on the T_P iterate level, following the characterization $\mathrm{LHM}(KB) = \bigcup_k T_P^k(\emptyset)$. At each level, EDB facts compile to variable nodes, and rule applications compile to AND/OR combinations of body BDDs obtained from the inductive hypothesis.

5.5 Proof Architecture

The complete verified chain:

$$\underbrace{\text{ProbLog program}}_{\texttt{ProbLogProgram}} \xrightarrow{\text{compilation}} \underbrace{\text{BDD}}_{\texttt{GroundBDDCompile}} \xrightarrow{\text{WMC}} \underbrace{P(q)}_{\texttt{bdd\_wmc\_correct}}$$

- **Soundness + Completeness**: the compiled BDD evaluates to true under assignment a if and only if the query holds in the corresponding ProbLog world.
- **WMC Correctness**: Weighted Model Counting on the BDD equals the distribution semantics weighted sum.
- **Therefore**: $\texttt{bdd\_wmc}(f, env) = P(q)$.

6 WM-PLN Subsumes ProbLog

The World Model – Probabilistic Logic Networks (WM-PLN) framework subsumes ProbLog: it computes the same probabilities and additionally provides confidence values and incremental evidence revision.

6.1 Subsumption: PLN Fuzzy-OR = ProbLog Noisy-OR

PLN's fuzzy-OR formula for independent causes is:

$$s_1 + s_2 - s_1 \cdot s_2 = 1 - (1 - s_1)(1 - s_2)$$

This is algebraically identical to the noisy-OR formula that arises from ProbLog's distribution semantics for OR-pattern programs (proved in Lean: `fuzzyOrMulti_eq_noisyOrMulti`, 0 `sorry`).

For the alarm example, both ProbMeTTa and PLN give $P(alarm) = 0.28$:

```
;; ProbMeTTa (BDD-based ProbLog):
!(?prob (alarm))                ;; => 0.28

;; PLN fuzzy-OR (native, no BDDs):
!(pln-fuzzy-or 0.1 0.2)         ;; => 0.28
```

6.2 Extension: Evidence-Based Reasoning with Confidence

ProbLog stores fixed-point probabilities. PLN stores *evidence*: pairs of positive and negative observation counts (n^+, n^-).

Definition 6.1 (Evidence and Simple Truth Value)**.** An evidence pair (n^+, n^-) yields:

$$\text{strength} = \frac{n^+}{n^+ + n^-} \tag{4}$$

$$\text{confidence} = \frac{n^+ + n^-}{n^+ + n^- + \kappa} \tag{5}$$

where κ is a prior context parameter.

Strength matches ProbLog's probability. Confidence measures how much evidence supports the estimate — something ProbLog cannot express.

6.3 Evidence Revision: What ProbLog Cannot Do

ProbLog stores $P(burglary) = 0.4$ as a weight. It cannot distinguish whether this came from 4 observations out of 10 (weak evidence) or 400 out of 1,000 (strong evidence). When new data arrives, ProbLog requires external relearning.

PLN stores the counts (n^+, n^-) and revises incrementally via *evidence aggregation* (`evidence-hplus`):

$$(n_1^+, n_1^-) \oplus (n_2^+, n_2^-) = (n_1^+ + n_2^+,\ n_1^- + n_2^-)$$

6.4 Demonstration: Convergence to Known Truth

We construct a synthetic benchmark with known ground truth ($P(alarm) = 0.28$) and reveal observations in three batches:

The estimate converges from 0.40 (initial small-sample error) to 0.2791 (within 0.001 of truth), while confidence rises from 0.50 to 0.91. ProbLog can match the final answer, but only via an external weight-relearning step — it has no internal revision mechanism.

After	P(alarm)	Error	Confidence	Method
10 windows	0.4000	0.1200	0.50	ProbLog (stuck) / WM-PLN
40 windows	0.3025	0.0225	0.80	WM-PLN revision
100 windows	0.2791	0.0009	0.91	WM-PLN revision
100 (rebuilt)	0.2791	0.0009	—	ProbLog (external relearning)

Table 3: Evidence revision converges toward truth with 133× error reduction. ProbLog matches the final answer only after a complete program rebuild.

The key insight: ProbLog represents $P = 0.4$ and cannot distinguish 4/10 from 400/1000. Those two cases should revise very differently when new data arrives, but ProbLog represents them identically. WM-PLN keeps the sufficient statistics (n^+, n^-), so the amount of revision is principled.

7 Getting ProbMeTTa Running on PeTTa

ProbMeTTa was originally developed for Hyperon Experimental MeTTa. Running it on PeTTa required fixing four compatibility issues:

1. `is-ground` **undefined**: PeTTa lacks this built-in. Fixed by wrapping SWI-Prolog's `ground/1` via a one-line Prolog predicate.
2. **Singleton list patterns**: PeTTa's `($h . $t)` cons pattern does not match singletons `(x)`. Fixed by using `car-atom`/`cdr-atom` instead of dotted-pair patterns.
3. `case` **branch ordering**: PeTTa's `case` with `once(match ...)` requires the `Empty` branch first.
4. `=>` **not evaluated**: PeTTa treats `=>` as structural, not as a reducible function. Fixed by introducing `prob-rule` as the rule declaration entrypoint.

After these fixes, all 27 ProbMeTTa tests pass on PeTTa.

8 Conclusion

We have shown three things:

1. **ProbMeTTa correctly implements ProbLog.** A formal proof in Lean 4 (1,160 lines, 0 `sorry`) covers BDD construction, Weighted Model Counting, and compilation soundness and completeness. Kernel-checked conformance tests link the abstract proof to the concrete MeTTa implementation. All 27 ProbMeTTa tests pass.
2. **WM-PLN subsumes ProbLog.** PLN's fuzzy-OR is algebraically identical to ProbLog's noisy-OR (proved: `fuzzyOrMulti_eq_noisyOrMulti`). For any ground definite clause ProbLog program, WM-PLN computes the same query probability (proved via BDD compilation soundness and completeness; negation-as-failure and annotated disjunctions additionally verified empirically via 27/27 ProbMeTTa tests).
3. **WM-PLN extends ProbLog with confidence and revision.** By storing evidence counts rather than point probabilities, WM-PLN provides confidence values and supports incremental revision. In a synthetic benchmark, evidence revision reduces estimation error by 133× (from 0.12 to 0.0009) while confidence rises from 0.50 to 0.91. ProbLog can match the final answer only via external relearning.

The key conceptual point: ProbLog discards the sufficient statistics behind its probability estimates. A weight of 0.4 could represent 4/10 or 400/1000 observations — ProbLog cannot tell the difference. WM-PLN retains the evidence (n^+, n^-), making revision principled and incremental.

Availability

- ProbMeTTa: `https://github.com/zariuq/ProbMeTTa`
- PeTTa: `https://github.com/zariuq/PeTTa` (branch `wmPLN`)
- Lean formalization: `Mettapedia/Logic/BDD/` in the Mettapedia project
- ProbLog (upstream): `https://github.com/ML-KULeuven/problog`

References

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